Journal of Research, Innovation and Technologies
https://www.acadlore.com/journals/JORIT

Technological, Organizational, and Environmental Drivers of E-Marketing Adoption and Marketing Performance: Evidence From Food-Processing MSMEs in Bali, Indonesia

Dewa Ayu Agung Shita Bhoraningrum*, Gede Mekse Korri Arisena

Faculty of Agriculture, Udayana University, 80114 Denpasar, Indonesia

* Correspondence: Dewa Ayu Agung Shita Bhoraningrum (sithabhoran@gmail.com)

Received: 07-16-2026 **Revised:** 08-30-2026 **Accepted:** 09-09-2026

Citation: Bhoraningrum, D. A. A. S. & Arisena, G. M. K. (2026). Technological, organizational, and environmental drivers of e-marketing adoption and marketing performance: Evidence from food-processing MSMEs in Bali, Indonesia. *J. Res. Innov. Technol.*, *5*(3), 283–297. https://doi.org/10.56578/jorit050304.

© 2026 by the author(s). Published by Acadlore Publishing Services Limited, Hong Kong. This article is available for free download and can be reused and cited, provided that the original published version is credited, under the CC BY 4.0 license.

Abstract: Electronic marketing (e-marketing) has become an important means for micro, small, and medium enterprises (MSMEs) to reach customers and manage marketing activities, yet its adoption remains uneven across small food businesses. The factors governing technology adoption and their relationships with marketing performance require further examination, particularly in tourism-dependent local markets. This study investigates the technological, organizational, and environmental determinants of e-marketing adoption and examines its mediating role in the marketing performance of animal-based food-processing MSMEs in Gianyar Regency, Bali, Indonesia. Survey data were collected from 94 business owners selected through purposive sampling and analyzed using partial least squares structural equation modeling (PLS-SEM) within the Technology–Organization–Environment (TOE) framework. E-marketing adoption was moderate among the surveyed enterprises. Technological factors showed the strongest association with adoption ($\beta = 0.432$), followed by organizational ($\beta = 0.372$) and environmental factors ($\beta = 0.333$). All three relationships were statistically significant ($p < 0.001$). E-marketing adoption was also positively associated with marketing performance ($\beta = 0.383$, $p < 0.001$). The estimated indirect effects through e-marketing adoption were 0.166, 0.143, and 0.128 for technological, organizational, and environmental factors, respectively, accounting for 28.96%–37.53% of their corresponding total effects. These findings indicate positive direct relationships between the three TOE dimensions and marketing performance, together with estimated indirect relationships through e-marketing adoption. The study extends the application of the TOE framework to a specific segment of the food-processing sector and provides empirical evidence for developing digital adoption strategies suited to small enterprises operating in tourism-oriented markets.

Keywords: E-marketing adoption; Technology–Organization–Environment framework; Marketing performance; Digital transformation; Food-processing MSMEs; Partial least squares structural equation modeling

JEL Classification: M31, O33, L26

1. Introduction

Traditional Balinese cuisine occupies an important position in the cultural and economic life of local communities. Beyond meeting everyday food needs, traditional dishes reflect established culinary practices, religious customs, and forms of social interaction that have developed over generations. Foods such as lawar, satay lilit, and ayam betutu remain closely associated with Balinese cultural identity and are frequently prepared for religious ceremonies, community gatherings, and other traditional occasions (Ardaba Kory & Sanica, 2022). The production and sale of these foods also support local livelihoods through micro, small, and medium enterprises (MSMEs), which connect traditional food preparation with commercial activities and employment opportunities (Mulyanto & Budi, 2025). In Bali, where tourism generates substantial demand for regional food products, culinary

MSMEs operate at the intersection of cultural preservation and market development. Their ability to maintain traditional products while responding to changing consumer preferences has become an important consideration in the management of local food businesses.

The adoption of electronic marketing (e-marketing) offers several possibilities for the continued development of these enterprises. Economically, digital channels allow small businesses to reach customers beyond their immediate locations, communicate product information at relatively low cost, and maintain contact with existing customers. Such capabilities are particularly relevant to food enterprises whose sales depend on both local residents and visitors. From a social perspective, online platforms provide additional means of presenting traditional Balinese foods to consumers who may be unfamiliar with their cultural origins. Photographs, short videos, customer reviews, and other forms of digital content can communicate product characteristics and preparation practices while retaining references to local culinary traditions. Digital promotion may also reduce the need for certain printed advertising materials, although its environmental implications depend on how digital activities are organized and used. These economic, social, and environmental considerations suggest that e-marketing should be examined not simply as an additional promotional channel, but as a form of technology adoption that may influence how traditional food enterprises organize marketing activities, maintain customer relationships, and respond to market conditions.

Gianyar Regency provides a relevant setting for examining these relationships. The regency is widely associated with Balinese arts, cultural tourism, and traditional food production. Animal-based processed foods, including babi guling, fish satay lilit, and ayam betutu, form part of its culinary offerings and are sold through businesses serving residents and tourists. The concentration of restaurants and related food establishments is particularly pronounced in areas with substantial tourism activity. According to the Central Statistics Agency of Gianyar Regency, Ubud District recorded 1,273 restaurants in 2023, compared with 305 in Sukawati District and 260 in Gianyar District. These figures indicate a considerable concentration of food-service activity in Ubud and Sukawati, where enterprises operate in markets shaped by visitor demand, local competition, and the growing use of online information when choosing food products and dining locations. For businesses in these areas, visibility on digital platforms may affect how potential customers discover products, compare alternatives, and establish contact with sellers. The local market therefore provides an opportunity to investigate the organizational and technological conditions associated with the adoption of digital marketing practices.

Digital technologies have changed the ways in which MSMEs communicate with customers and conduct promotional activities. E-marketing allows businesses to distribute product information, reach geographically dispersed consumers, and maintain communication through online channels without relying exclusively on conventional advertising methods (Astuti et al., 2023). For food enterprises, visual presentation is particularly relevant because purchasing decisions may be influenced by product appearance, preparation methods, and information available before a customer visits a business or places an order. Photographs, videos, and other forms of online content can therefore contribute to the presentation of culinary products and the communication of their distinctive characteristics (Meranti, 2020). Previous research has also associated the consistent use of digital marketing practices with greater product visibility and improved marketing performance among MSMEs (Zainarti et al., 2024). Nevertheless, the availability of digital platforms does not necessarily lead to their systematic use. Enterprises may establish social media accounts or register with online marketplaces while making only limited use of these channels for customer interaction, promotional planning, or market development. The distinction between access to digital technology and its effective adoption is therefore important when examining marketing practices in small food businesses.

Despite the increasing availability of digital marketing tools, MSMEs continue to encounter difficulties in incorporating these technologies into routine business operations. Limited digital literacy, shortages of personnel with relevant skills, and inadequate organizational preparation may restrict the extent to which online platforms are used for marketing purposes (Aqham et al., 2024). These constraints indicate that e-marketing adoption cannot be explained solely by the technical availability of digital services. Earlier studies have identified technological, organizational, and environmental conditions as relevant determinants of digital marketing adoption among MSMEs (Priyanto et al., 2020). However, the application of these findings to particular business settings requires further examination. Much of the existing discussion addresses MSMEs as a broad category or considers digital commerce and business transformation without distinguishing the operating conditions of individual industries. Animal-based food-processing enterprises represent a more specific setting in which product characteristics, local culinary traditions, customer expectations, and tourism-related demand may shape the use of digital marketing channels. A further issue concerns the relationship between the conditions supporting adoption and the marketing outcomes associated with it. Identifying the determinants of technology adoption does not, by itself, explain whether those determinants are associated with marketing performance directly or through the actual use of digital marketing practices. This distinction provides the basis for examining e-marketing adoption as an intervening mechanism within the relationships between technological, organizational, and environmental conditions and marketing performance.

Against this background, the present study investigates e-marketing adoption and marketing performance among MSMEs producing animal-based processed food products in Gianyar Regency, Bali, Indonesia. Three related objectives are addressed. First, the study describes the extent of e-marketing adoption among the surveyed enterprises. Second, it examines the relationships between technological, organizational, and environmental conditions and the adoption of e-marketing. Third, it assesses the relationship between e-marketing adoption and marketing performance, including the mediating role of adoption in the associations between the three contextual dimensions and performance outcomes. The analysis is organized around the Technology–Organization–Environment (TOE) framework, which provides a basis for examining technology adoption in relation to the characteristics of the technology, the internal conditions of an organization, and the external environment in which it operates. In the present study, the technological context encompasses perceived usefulness, ease of use, compatibility, accessibility, and cost efficiency. The organizational context concerns management support, employee capabilities, previous digital experience, infrastructure readiness, and organizational arrangements. The environmental context includes competitive pressure, customer demand, community support, government assistance, and access to training. These dimensions are examined in relation to e-marketing adoption, represented by the use of social media, online marketplaces, digital content, customer interaction, and technology integration. Marketing performance is considered in terms of sales growth, customer development, market expansion, promotional effectiveness, customer loyalty, and brand equity.

The relationships proposed in this study follow the view that the adoption of e-marketing depends on both the capacity of enterprises to use digital tools and the conditions under which those tools are introduced into business activities. Technological characteristics may influence adoption by determining whether digital platforms are accessible, compatible with existing practices, and sufficiently useful for routine marketing tasks. Organizational conditions may affect whether businesses possess the managerial commitment, skills, and infrastructure needed to maintain these activities. Environmental conditions may provide additional incentives or constraints through competition, changing customer expectations, institutional assistance, and training opportunities. Once adopted, digital marketing practices may be associated with marketing performance through wider customer access, more frequent communication, and the coordination of promotional activities across digital channels. The TOE dimensions may also have direct relationships with marketing performance because technological resources, organizational capabilities, and external market conditions can affect business operations beyond their role in e-marketing adoption. Accordingly, the empirical model examines the hypothesized relationships involving e-marketing adoption, together with the direct associations between the three TOE dimensions and marketing performance required to assess the proposed mediation pathways.

On this basis, the following hypotheses are proposed:

H1: The technological context has a positive effect on e-marketing adoption among MSMEs producing animal-based processed food products in Gianyar Regency.

H2: The organizational context has a positive effect on e-marketing adoption among MSMEs producing animal-based processed food products in Gianyar Regency.

H3: The environmental context has a positive effect on e-marketing adoption among MSMEs producing animal-based processed food products in Gianyar Regency.

H4: E-marketing adoption has a positive effect on marketing performance among MSMEs producing animal-based processed food products in Gianyar Regency.

H5: E-marketing adoption mediates the relationship between the technological context and marketing performance among MSMEs producing animal-based processed food products in Gianyar Regency.

H6: E-marketing adoption mediates the relationship between the organizational context and marketing performance among MSMEs producing animal-based processed food products in Gianyar Regency.

H7: E-marketing adoption mediates the relationship between the environmental context and marketing performance among MSMEs producing animal-based processed food products in Gianyar Regency.

By examining these relationships within a specific segment of the Balinese food industry, the study contributes to the empirical application of the TOE framework in two respects. First, it considers technology adoption among small enterprises whose marketing activities are closely connected to local culinary products and tourism-related demand, rather than treating MSMEs as a uniform business category. Second, it examines e-marketing adoption alongside direct and indirect relationships between contextual conditions and marketing performance, allowing the role of actual digital marketing practices to be distinguished from that of the conditions associated with their adoption. The findings provide a basis for understanding differences in digital marketing adoption among the surveyed enterprises and for identifying areas in which technological access, organizational preparation, and external support may be relevant to the development of their marketing activities.

2. Methodology

This study employed a quantitative, cross-sectional survey design to examine the relationships between technological, organizational, and environmental conditions, e-marketing adoption, and marketing performance

among MSMEs producing animal-based processed food products. The research was conducted in Gianyar Regency, Bali Province, Indonesia, with particular attention to Ubud and Sukawati Districts. These districts were purposively selected because of their concentration of food-service establishments and their relevance to the local tourism economy. According to the 2023 records of the Central Statistics Agency of Gianyar Regency, Ubud and Sukawati had the largest numbers of restaurants among the districts considered in the study. Their selection provided an opportunity to examine e-marketing practices in areas where food businesses operate alongside substantial tourism-related commercial activity. The study was conducted between January and March 2026, during which research preparation, data collection, data analysis, and initial reporting were undertaken.

The investigation drew primarily on quantitative questionnaire data, supplemented by brief interviews and secondary information. Structured questionnaires were administered to owners of MSMEs engaged in the production and sale of animal-based processed food products. The questionnaire data provided the basis for measuring the study constructs and estimating the relationships specified in the research model. Brief interviews were also conducted to obtain contextual information concerning the circumstances under which respondents adopted and used digital marketing tools. These interviews served a supporting role and were not treated as a separate qualitative dataset for statistical modeling. Secondary information was obtained from the Central Statistics Agency of Gianyar Regency, the local Cooperatives and MSMEs Office, and relevant academic publications. These sources were used to establish the study context and inform the selection of the research area and measurement constructs.

The reference population was drawn from the 2023 records of food-service establishments in Ubud and Sukawati Districts, which listed a combined total of 1,578 businesses. An initial sample-size estimate was obtained using Slovin's formula with a specified margin of error of 10%. On this basis, 94 respondents were included in the study. The sample was distributed proportionally between the two districts, comprising 76 MSMEs from Ubud and 18 from Sukawati. This allocation reflected the relative concentration of establishments in the two study areas. The sample-size calculation was used as a practical reference for determining the number of respondents; it did not establish a probability-based margin of sampling error because the final respondents were selected through purposive rather than random sampling.

Respondents were selected according to criteria intended to identify enterprises with direct experience of e-marketing. Eligible businesses were required to produce or sell animal-based processed food products, operate within Ubud or Sukawati District, and use digital platforms for marketing or promotional activities. The platforms considered included Google Maps, GoFood, GrabFood, and ShopeeFood, which provide different forms of online business visibility, customer access, and food-ordering services. Businesses were also identified through food-service categories and platform listings associated with popular or frequently purchased products. These criteria ensured that respondents could provide information about their use of digital marketing tools and the organizational conditions surrounding their adoption. However, because enterprises without digital platform experience were not included, the sample was not intended to represent the full range of adoption behavior among all food-related MSMEs in Gianyar Regency. The reported level of e-marketing adoption therefore refers specifically to the enterprises included in the survey.

The research model comprised three exogenous constructs, one mediating construct, and one outcome construct. The exogenous constructs were technological context (X_1), organizational context (X_2), and environmental context (X_3), derived from the TOE framework and operationalized with reference to Priyanto et al. (2020). E-marketing adoption (Z) was specified as the mediating construct, while marketing performance (Y) was treated as the outcome construct. The technological context represented the perceived characteristics and accessibility of digital marketing technologies, including ease of use, perceived usefulness, system compatibility, accessibility, and cost efficiency. The organizational context covered internal conditions relevant to technology use, including management support, human resource capability, digital experience, infrastructure readiness, and organizational structure. The environmental context represented external influences, including competitive pressure, customer demand, community support, government support, and access to training.

E-marketing adoption was assessed through indicators covering social media utilization, online marketplace utilization, content intensity, customer interaction, and technology integration. These indicators were intended to capture the extent to which digital marketing practices had been incorporated into the enterprises' promotional and customer-related activities. Marketing performance was measured using indicators relating to sales growth, customer loyalty, customer growth, market expansion, promotional effectiveness, and brand equity. The measurement indicators were adapted from the studies identified in Table 1. All items were assessed using a five-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree). The constructs, their corresponding indicators, and the supporting literature are presented in Table 1.

Data analysis consisted of descriptive analysis and partial least squares structural equation modeling (PLS-SEM). Descriptive statistics were used to summarize respondent characteristics and describe the reported extent of e-marketing adoption. PLS-SEM was applied using SmartPLS 4.0 (statistical software package) to estimate the relationships among the TOE dimensions, e-marketing adoption, and marketing performance. The structural model included paths from the three TOE dimensions to e-marketing adoption, a path from e-marketing adoption to

marketing performance, and direct paths from the TOE dimensions to marketing performance. The latter paths were retained to examine whether the contextual factors were associated with marketing performance independently of their relationships with e-marketing adoption. This specification also provided the basis for estimating the indirect relationships proposed in the mediation hypotheses.

Table 1. Measurement constructs, indicators, and references

Construct	Indicators	Reference
Technology context (X_1)	Ease of use; perceived usefulness; system compatibility; accessibility; cost efficiency	Jadhav et al. (2023); Priyanto et al. (2020); Svoboda (2024);
Organization context (X_2)	Management support; human resource capability; digital experience; infrastructure readiness; organizational structure	Mulyanto & Budi (2025); Priyanto et al. (2020)
Environment context (X_3)	Competitive pressure; customer demand; community support; government support; training accessibility	Ardaba Kory & Sanica (2022); Priyanto et al. (2020)
E-marketing adoption (Z)	Social media utilization; marketplace utilization; content intensity; customer interaction; technology integration	Jadhav et al. (2023); Priyanto et al. (2020); Safitri et al. (2022)
Marketing performance (Y)	Sales growth; customer loyalty; customer growth; market expansion; promotional effectiveness; brand equity	Nofrisel et al. (2023); Supriyati et al. (2024); Tuluki et al. (2023)

The measurement model was assessed using indicator loadings, average variance extracted (AVE), composite reliability, and Cronbach's alpha. Indicator loadings were examined to evaluate the contribution of individual items to their respective constructs, while AVE was used to assess convergent validity. Composite reliability and Cronbach's alpha were used to evaluate internal consistency. The structural model was examined through path coefficients and their associated statistical tests. Variance inflation factors (VIFs) were considered in assessing collinearity among predictor constructs, while the coefficient of determination (R^2), predictive relevance (Q^2), and effect size (f^2) were included among the model assessment measures where applicable. The statistical significance of the estimated relationships was evaluated using the bootstrapping procedure implemented in SmartPLS 4.0, with reference to the resulting t-statistics and p-values.

The mediating role of e-marketing adoption was examined by estimating the indirect relationships between each TOE dimension and marketing performance through e-marketing adoption. For each relationship, the indirect effect was calculated from the relevant constituent path coefficients, while the total effect was obtained by adding the direct and indirect effects. Variance accounted for (VAF) was calculated as the ratio of the indirect effect to the corresponding total effect and expressed as a percentage. This measure was used to describe the relative contribution of the indirect pathway to the overall estimated relationship. The statistical interpretation of mediation, however, depends on the significance and uncertainty of the indirect effect rather than on the VAF percentage alone. Accordingly, the mediation results were considered in relation to the estimated direct and indirect paths, with the limitations of the available statistical evidence taken into account.

The overall analytical procedure, including the description of respondents, assessment of the measurement and structural models, and examination of the direct and indirect relationships, is summarized in Figure 1.

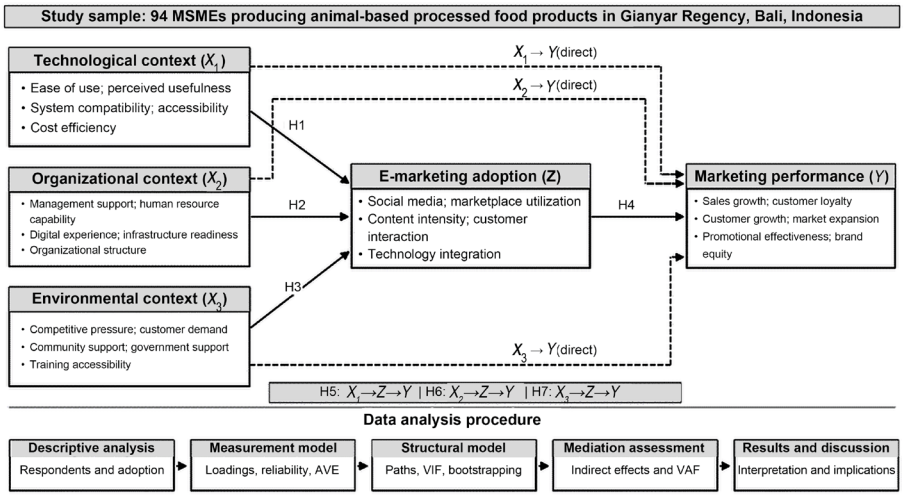


Figure 1. Research framework and data analysis procedure

Note: MSMEs = micro, small, and medium enterprises; AVE = average variance extracted; VIF = variance inflation factor; VAF = variance accounted for. X_1, X_2, X_3 represent technological, organizational, and environmental contexts, respectively; Z = e-marketing adoption; Y = marketing performance. Solid arrows indicate hypothesized direct paths (H1–H4), while dashed arrows represent the direct effects of TOE dimensions on marketing performance included for mediation testing.

3. Results and Discussion

3.1 Characteristics of Respondents

The survey included 94 owners of MSMEs producing animal-based processed food products in Gianyar Regency, Bali. All participating enterprises had experience using digital platforms for marketing or promotional activities, consistent with the sampling criteria described in Section 2. Respondent characteristics were examined to establish the business and demographic context in which e-marketing adoption occurred. The descriptive analysis covered gender, age, business location, educational attainment, length of business operation, number of employees, and annual revenue. The distributions are presented in Table 2.

Table 2. Demographic and business characteristics of the surveyed MSMEs in Gianyar Regency

No.	Characteristics	Category	Percentage (%)
1	Gender	Women	53.2
		Male	46.8
2	Age	21–40 years old	63.8
		56–74 years old	4.3
		Other/not reported	31.9
3	District	Ubud	80.9
		Sukawati	19.1
4	Educational attainment	Senior high school	48.9
		Bachelor's degree	29.8
		Elementary/junior high school	4.3
		Other/not reported	17
5	Length of business operation	>3.5 years	79.8
		≤3.5 years	20.2
6	Number of employees	5–19 people	68.1
		20–99 people	2.1
		Other/not reported	29.8
7	Revenue per year	IDR 300 million–IDR 2.5 billion	67
		IDR 2.5 billion–IDR 50 billion	3.2
		Other/not reported	29.8

Note: MSMEs = micro, small, and medium enterprises; IDR = Indonesian rupiah. Percentages are based on the 94 surveyed enterprises. Categories labeled "Other/not reported" follow the classifications in the original survey summary.

Women accounted for 53.2% of respondents, compared with 46.8% for men. The relatively balanced gender distribution indicates that both women and men were substantially represented among the owners of the surveyed food enterprises. Women's slightly higher representation is consistent with their participation in the management and commercial activities of local culinary businesses. However, the distribution alone does not establish differences in digital marketing behavior between female and male owners. Gender is therefore treated as a descriptive characteristic rather than an explanatory factor in the structural model.

Regarding age, 63.8% of respondents were between 21 and 40 years old, while 4.3% were between 56 and 74 years old. The remaining 31.9% were classified under other or unreported age categories. The concentration of respondents in the younger age group is relevant to the study because digital marketing often requires familiarity with mobile applications, online communication, and platform-based business services. Nevertheless, age should not be interpreted as a direct indicator of digital competence. The extent to which individual owners can use digital technologies depends on their experience, training, and access to appropriate resources, none of which can be established from the age distribution alone.

The geographical distribution reflected the sampling allocation, with 80.9% of enterprises located in Ubud and 19.1% in Sukawati. This concentration was consistent with the larger number of food-service establishments recorded in Ubud. As a major tourism destination, Ubud provides a setting in which restaurants and food producers serve both local consumers and visitors who may rely on digital platforms when searching for products and dining options. Such market conditions make online visibility and customer communication particularly relevant to business operations. At the same time, the greater representation of Ubud enterprises means that the aggregate findings are more strongly informed by businesses operating in that district.

Educational attainment varied across the respondents. Senior high school graduates accounted for 48.9% of the sample, followed by bachelor's degree holders at 29.8%. Respondents with elementary or junior high school education represented 4.3%, while 17.0% were recorded in other or unreported categories. The distribution shows that the surveyed enterprises were managed by individuals with different formal educational backgrounds. Their participation in digital marketing activities suggests that the use of online platforms is not confined to university-educated business owners. However, the data do not establish whether educational attainment was associated with

the intensity or effectiveness of e-marketing adoption. Such relationships would require separate statistical examination.

Most enterprises had operated for more than 3.5 years, accounting for 79.8% of the sample, while 20.2% had been established for 3.5 years or less. This indicates that the survey primarily covered businesses with several years of operating experience rather than newly established enterprises. Longer business operation may provide owners with greater familiarity with customer preferences, seasonal demand, and local competition. It may also influence decisions about whether existing marketing practices should be maintained or supplemented with digital channels. Nevertheless, business age was not included as a predictor in the structural model, and its relationship with e-marketing adoption cannot be inferred directly from the descriptive results.

In terms of employment, 68.1% of enterprises reported having 5–19 employees, while 2.1% employed 20–99 workers. The remaining 29.8% were classified under other or unreported employment categories. These figures indicate that enterprises with relatively small workforces formed the largest identifiable group in the sample. For such businesses, marketing responsibilities may be shared among owners and employees rather than assigned to specialized departments. This operating structure makes organizational readiness relevant, particularly where digital marketing requires regular content preparation, customer responses, and coordination with online ordering platforms.

Annual revenue showed a similar concentration. Approximately 67.0% of enterprises reported revenue between IDR 300 million and IDR 2.5 billion, while 3.2% reported revenue between IDR 2.5 billion and IDR 50 billion. Other or unreported revenue categories accounted for 29.8%. The distribution suggests variation in the commercial scale of the participating enterprises, although the available categories do not support a detailed assessment of their financial capacity. Revenue should therefore be regarded as contextual information rather than direct evidence of business maturity or marketing effectiveness.

Taken together, the descriptive findings indicate that the study covered predominantly established food enterprises, with substantial representation from Ubud and from businesses employing relatively small workforces. These characteristics provide context for interpreting the subsequent structural relationships. In particular, the combination of tourism-related market exposure, existing operating experience, and limited organizational scale may influence how digital technologies are incorporated into marketing activities. Because all respondents were selected from enterprises already using digital platforms, the results describe variation within an existing user population rather than the adoption rate among all food-related MSMEs in Gianyar Regency.

3.2 Relationships Between the Technology–Organization–Environment Dimensions, E-Marketing Adoption, and Marketing Performance

Marketing performance was examined as an outcome associated with the technological, organizational, and environmental conditions of the surveyed enterprises and their adoption of e-marketing. In the present study, marketing performance encompassed sales growth, customer development, customer loyalty, market expansion, promotional effectiveness, and brand equity. These dimensions reflect the extent to which enterprises achieve their marketing objectives, consistent with the broader understanding of marketing performance discussed by Kotler & Keller (2016). The structural model distinguished the direct relationships of the three TOE dimensions with marketing performance from the relationship between e-marketing adoption and marketing performance. This distinction is relevant because access to technology, internal business capabilities, and external market conditions may be associated with marketing outcomes even when their relationships are not fully explained by the adoption of digital marketing practices. Chaffey & Ellis-Chadwick (2019) similarly discussed the role of digital marketing in extending customer reach and supporting communication through online channels. The estimated direct relationships are reported in Table 3.

Table 3. Estimated direct relationships with marketing performance

Relationships Between Variables	Path Coefficients (β)	*t*-Statistic	*p*-Value	Decision
Technology → marketing performance	0.287	4.239	<0.001	Supported
Organization → marketing performance	0.238	3.717	<0.001	Supported
Environment → marketing performance	0.314	4.249	<0.001	Supported
Marketing adoption → marketing performance	0.383	4.040	<0.001	Supported

Note: β denotes the estimated path coefficient. The reported significance values are taken from the original structural model results.

All four estimated paths were positive and statistically significant, with *p*-values below 0.001. E-marketing adoption had the largest standardized path coefficient (β = 0.383), followed by the environmental context (β = 0.314), technological context (β = 0.287), and organizational context (β = 0.238). These estimates indicate that e-marketing adoption had the strongest direct association with marketing performance among the four predictors included in the model. The relative magnitudes describe the estimated relationships within the surveyed sample

and should not be interpreted as proof that changes in one factor would produce equivalent changes in business performance.

The coefficient for e-marketing adoption suggests that enterprises reporting more extensive use of digital marketing practices also tended to report stronger marketing performance after accounting for the other predictors in the model. This relationship is relevant to businesses that use social media, online marketplaces, and food-ordering applications to present products and communicate with customers. Maintaining updated product information, responding to customer inquiries, and coordinating promotional activities across platforms may be associated with more consistent market engagement. The result therefore places emphasis on the actual incorporation of digital tools into marketing routines, rather than on their availability alone. However, the cross-sectional design does not establish whether greater adoption preceded improved performance or whether better-performing enterprises were more capable of investing in digital marketing.

The technological context showed a positive relationship with marketing performance ($\beta = 0.287$, $p < 0.001$). This finding indicates that enterprises reporting more favorable technological conditions also tended to report stronger marketing outcomes. Ease of use, accessibility, compatibility with existing business practices, and perceived cost efficiency may affect the practical value of digital tools for small enterprises. For example, platforms that can be operated without extensive technical support may allow owners to maintain product information and customer communication alongside other business responsibilities. Fifian (2025) discussed the relationship between digital technology use and marketing activities among MSMEs, while Hakim et al. (2025) examined digital marketing practices in culinary enterprises. The present result is consistent with the broader observation that technological conditions are relevant to the marketing activities of small businesses, although it does not identify which individual technological indicator contributed most strongly to the estimated relationship.

Organizational context was also positively associated with marketing performance ($\beta = 0.238$, $p < 0.001$), although its coefficient was the smallest among the four predictors. This finding suggests that management support, employee capabilities, digital experience, and infrastructure readiness remained relevant after technological and environmental conditions and e-marketing adoption were considered. For small food enterprises, the ability to maintain digital marketing activities may depend on how responsibilities are assigned, whether employees can operate online platforms, and whether owners provide the time and resources needed for regular customer communication. The findings reported by Hakim et al. (2025) and Putri & Arifiani (2022) provide related evidence concerning organizational conditions and marketing outcomes. In the present study, the organizational coefficient indicates a positive relationship, but its smaller magnitude should not be taken to mean that internal capabilities are unimportant. Their contribution may also be reflected in the indirect relationship operating through e-marketing adoption, examined in Section 3.4.

The environmental context had the second-largest direct coefficient ($\beta = 0.314$, $p < 0.001$). External conditions, including competition, customer expectations, community support, government assistance, and training opportunities, were therefore positively associated with marketing performance in the estimated model. This relationship is particularly relevant to enterprises operating in tourism-oriented food markets, where consumer demand and competitive conditions may change across locations and periods. Businesses that are more responsive to these conditions may also be better positioned to communicate with customers and adjust their marketing activities. Mulyanto & Budi (2025), Nuraini & Mardikaningsih (2024), and Rahman & Riofita (2024) discussed related relationships between business environments and enterprise performance. The present findings extend this discussion to a specific group of animal-based food-processing MSMEs, while remaining limited to the associations observed in the survey.

Overall, the direct-effect results indicate that marketing performance was associated with both the contextual conditions of the enterprises and their reported use of e-marketing. The larger coefficient for adoption suggests that digital marketing practices deserve particular attention when interpreting these relationships. At the same time, the significant direct coefficients for the three TOE dimensions indicate that the relationships between enterprise conditions and marketing performance were not confined to the adoption pathway. The next section examines how those conditions were associated with e-marketing adoption itself.

3.3 Technological, Organizational, and Environmental Determinants of E-Marketing Adoption

The adoption of e-marketing depends on more than the availability of digital platforms. Enterprises must also determine whether these platforms are suitable for their operations, whether sufficient internal resources are available to use them, and whether external market conditions justify their continued use. Within the TOE framework, these considerations are represented by technological, organizational, and environmental contexts. For the surveyed food enterprises, adoption included the use of social media, online marketplaces, digital promotional content, customer interaction tools, and the integration of digital services into marketing activities. Chaffey & Ellis-Chadwick (2019) described digital marketing as a means of organizing communication and reaching customers through online channels. In the present analysis, attention was directed to the conditions associated with the extent to which such practices had been adopted.

The estimated relationships between the three TOE dimensions and e-marketing adoption are presented in Table 4.

Table 4. Estimated relationships between the TOE dimensions and e-marketing adoption

Relationships Between Variables	Path Coefficients (β)	*t*-Statistic	*p*-Value	Decision
Technology → e-marketing adoption	0.432	6.093	<0.001	Supported
Organization → e-marketing adoption	0.372	5.292	<0.001	Supported
Environment → e-marketing adoption	0.333	3.845	<0.001	Supported

Note: TOE = Technology–Organization–Environment.

All three paths were positive and statistically significant ($p < 0.001$). The technological context had the largest standardized coefficient ($\beta = 0.432$), followed by organizational context ($\beta = 0.372$) and environmental context ($\beta = 0.333$). The findings supported H1, H2, and H3, indicating that each TOE dimension was positively associated with e-marketing adoption among the surveyed enterprises. The differences in coefficient magnitude suggest that technological conditions were particularly prominent in the estimated model. However, the coefficients alone do not establish that the differences between the three relationships were statistically significant.

The technological coefficient indicates that enterprises reporting more favorable technological conditions also reported more extensive e-marketing adoption. This relationship is consistent with the practical requirements of digital marketing in small food businesses. Owners and employees need to access platforms, upload product information, respond to messages, and manage customer interactions while carrying out routine production and sales activities. Technologies that are perceived as accessible, compatible with existing operations, and manageable in terms of cost are more likely to be incorporated into these activities. Andika et al. (2021) and Wahyudin et al. (2023) also examined technological readiness and digital marketing adoption among Indonesian enterprises. The present result provides additional evidence from animal-based food-processing businesses operating in a tourism-oriented local market.

The relatively large technological coefficient can also be considered in relation to the commercial characteristics of Gianyar Regency. Consumers searching for local food products may encounter businesses through Google Maps, Instagram, TikTok, and online food-delivery applications before making purchasing decisions. For enterprises operating in this environment, the ability to maintain accurate online information and communicate through commonly used platforms may be closely connected to their marketing activities. Reliable access to digital devices and internet services, together with the ability to operate relevant applications, can make these activities easier to maintain. Nevertheless, the study did not directly measure consumer search behavior or establish that organizational and environmental conditions were uniform across the surveyed businesses. The local tourism context therefore offers a plausible interpretation of the technological relationship rather than a separately tested explanation for its magnitude.

The organizational context also showed a positive association with e-marketing adoption ($\beta = 0.372$, $p < 0.001$). This result indicates that the use of digital marketing was related to the internal capacity of enterprises to organize and maintain technology-based activities. Management support may affect whether time and resources are allocated to online promotion, while employee capabilities and previous digital experience may influence how consistently platforms are used. Infrastructure readiness and organizational arrangements may also determine whether customer inquiries, online orders, and promotional content can be managed alongside routine business operations. Kaban et al. (2024) and Wahyudin et al. (2023) reported related findings concerning organizational readiness and digital marketing practices. The positive coefficient observed here suggests that technology adoption in small food enterprises is associated not only with the characteristics of digital tools but also with the conditions under which they are used.

From a managerial perspective, the organizational findings point to the importance of assigning clear responsibilities for digital marketing activities. Small enterprises may not require specialized marketing departments, but regular platform maintenance, timely customer responses, and coordination between online promotion and product availability still require organizational attention. Training may be useful where owners or employees lack experience with particular applications, especially those involving online ordering, promotional content, and customer communication. Management commitment is equally relevant because the continued use of digital platforms requires time, personnel, and coordination with routine business operations. For food-processing MSMEs with limited staff and financial resources, these findings suggest that organizational readiness is closely related to the ability to incorporate e-marketing into established business practices. Nevertheless, internal capabilities do not operate independently of the market conditions surrounding these enterprises. Competitive pressure, changing customer expectations, and access to external support may also shape the extent to which digital marketing practices are adopted.

The environmental context was positively associated with e-marketing adoption ($\beta = 0.333$, $p < 0.001$), indicating that external market conditions remained relevant alongside technological and organizational factors. Although its standardized coefficient was smaller than those of the technological and organizational contexts, the relationship remained statistically significant. This finding suggests that external market conditions were relevant

to the extent of digital marketing adoption among the surveyed enterprises. Competitive pressure and changing customer expectations may encourage food businesses to maintain an online presence, while government assistance, community support, and access to training may influence their capacity to use digital platforms. In tourism-oriented markets, these conditions are particularly relevant because enterprises serve customers with different purchasing habits, information needs, and expectations regarding online accessibility. The environmental relationship therefore provides a further explanation of why digital marketing adoption cannot be understood solely in terms of technological availability or internal organizational resources.

The practical implications of this relationship extend beyond individual enterprises. Government agencies, universities, business associations, and digital platform providers may contribute to digital adoption by making technical guidance and training more accessible to small food businesses. Such assistance is particularly relevant where owners have limited experience with platform management or lack the resources to obtain specialized marketing services. Training that addresses routine activities, including the preparation of product information, management of online listings, and communication with customers, may be more immediately applicable than general instruction in digital technology. Cooperation between local institutions and enterprises may also help identify difficulties associated with platform access and changing customer requirements. These measures should nevertheless be regarded as practical implications of the observed associations rather than interventions whose effectiveness was directly tested in the present study.

Considered together, the three TOE dimensions were positively associated with e-marketing adoption, although their estimated relationships differed in magnitude. Technological conditions showed the largest coefficient, indicating that the accessibility and suitability of digital tools were particularly prominent among the surveyed enterprises. Organizational readiness was also relevant, suggesting that internal capabilities were associated with the continued use of those tools. Environmental conditions provided an additional dimension through competition, customer expectations, and external support. The findings therefore suggest that e-marketing adoption among food-processing MSMEs is associated with a combination of technological resources, organizational arrangements, and market circumstances. The extent to which adoption accounts for the relationships between these contextual conditions and marketing performance is examined in the following section.

3.4 Mediating Role of E-Marketing Adoption in the Relationships Between the Technology–Organization–Environment Dimensions and Marketing Performance

The preceding analyses identified two sets of relationships. First, technological, organizational, and environmental conditions were positively associated with e-marketing adoption. Second, these contextual conditions and e-marketing adoption were each positively associated with marketing performance. Considered separately, these results establish the estimated direct relationships within the structural model. They do not, however, indicate how much of the relationship between each TOE dimension and marketing performance is represented by the pathway through e-marketing adoption. The mediation analysis was therefore used to examine the indirect relationships specified in H5–H7 and to distinguish them from the corresponding direct paths.

E-marketing adoption was specified as an intervening construct because the presence of favorable technological or organizational conditions does not necessarily mean that digital marketing tools are incorporated into routine business activities. For example, an enterprise may have access to suitable applications and possess employees capable of operating them, but the extent to which these resources are used for customer communication and promotion may vary. Similarly, competitive pressure or institutional support may be associated with marketing performance through several mechanisms, one of which is the adoption of digital platforms. Chaffey & Ellis-Chadwick (2019) discussed the role of digital marketing in customer communication and market access, providing a basis for considering adoption as a possible pathway connecting enterprise conditions with marketing outcomes. In the present study, the indirect effects were estimated from the relevant path coefficients, and VAF was calculated to describe their contribution to the corresponding total effects.

Table 5. Estimated direct, indirect, and total relationships with marketing performance through e-marketing adoption

Relationship	Direct Effect (β)	Indirect Effect (β)	Total Effect (β)	VAF
Technology → e-marketing adoption → marketing performance	0.287	0.166	0.453	36.64%
Organization → e-marketing adoption → marketing performance	0.238	0.143	0.381	37.53%
Environment → e-marketing adoption→ marketing performance	0.314	0.128	0.442	28.96%

Note: β denotes the estimated path coefficient; VAF = variance accounted for, calculated as the indirect effect divided by the total effect and expressed as a percentage. The indirect effects are reported as estimated values. Their statistical significance cannot be established from VAF alone and requires the corresponding bootstrap test results.

As shown in Table 5, all three estimated indirect effects were positive. The technological context had a direct effect of 0.287 and an indirect effect of 0.166 through e-marketing adoption, resulting in a total effect of 0.453. The indirect pathway accounted for 36.64% of the estimated total relationship. For organizational context, the direct and indirect effects were 0.238 and 0.143, respectively, giving a total effect of 0.381 and a VAF of 37.53%. The environmental context had a direct effect of 0.314 and an indirect effect of 0.128, with a total effect of 0.442 and a VAF of 28.96%. These estimates indicate that the indirect pathway through e-marketing adoption represented a substantial proportion of each total relationship, while the direct pathways remained positive. However, the classification of these relationships as statistically significant partial mediation requires confirmation from bootstrap confidence intervals or equivalent significance tests for the indirect effects.

The technological pathway had the largest estimated indirect effect ($\beta = 0.166$). This result is consistent with the relatively strong relationship between technological context and e-marketing adoption reported in Section 3.3. Enterprises with more favorable technological conditions tended to report greater adoption of digital marketing practices, which was in turn associated with stronger marketing performance. The estimated pathway suggests that the relationship between technological readiness and marketing outcomes may be partly represented by the extent to which digital tools are used in business activities. Accessible applications, compatible systems, and manageable operating costs may be relevant to the regular use of online promotion and customer communication. Nevertheless, the direct effect of technological context on marketing performance remained positive ($\beta = 0.287$), indicating that the estimated relationship was not confined to the adoption pathway.

This distinction is important for interpreting the role of technology in small food enterprises. Access to digital platforms may be associated with business performance through several activities, including the management of customer information, communication with suppliers, and coordination of sales processes. E-marketing adoption represents a more specific set of activities concerned with promotion, market access, and customer interaction. The findings reported by Andika et al. (2021) and Wahyudin et al. (2023) provide related evidence concerning technological readiness and digital marketing adoption among Indonesian MSMEs. The present analysis adds a distinction between the direct relationship of technological conditions with marketing performance and the estimated indirect relationship through actual e-marketing practices. For enterprises with limited resources, this distinction suggests that access to technology and its incorporation into routine marketing activities should be considered separately when identifying areas for managerial attention.

The organizational context showed an estimated indirect effect of 0.143, representing 37.53% of its total effect on marketing performance. Although this indirect coefficient was smaller than that of the technological context, its VAF was the largest among the three TOE dimensions. This difference indicates that the pathway through e-marketing adoption accounted for a relatively greater share of the estimated organizational relationship with marketing performance. Management support, employee capabilities, digital experience, and infrastructure readiness may be associated with how consistently enterprises use digital platforms for promotion and customer communication. Where these internal conditions are more favorable, the use of e-marketing may be more closely integrated with everyday business activities.

At the same time, the direct organizational effect remained positive ($\beta = 0.238$), suggesting that organizational conditions were associated with marketing performance beyond the estimated adoption pathway. This is reasonable in the context of small food businesses, where managerial decisions and employee capabilities are relevant to several activities that are not limited to digital promotion. The findings discussed by Kaban et al. (2024) similarly point to the relevance of organizational readiness in digital marketing practices among food-related MSMEs. In the present sample, the results suggest that organizational arrangements deserve attention not only because they are associated with technology adoption, but also because they are related to the broader management of marketing activities. The estimated indirect relationship provides a basis for examining whether more consistent digital marketing practices represent one mechanism through which internal capabilities are associated with marketing outcomes.

The environmental context had the smallest estimated indirect effect ($\beta = 0.128$), accounting for 28.96% of its total relationship with marketing performance. Its direct effect was 0.314, indicating that a larger proportion of the estimated total relationship was represented by the direct pathway. This pattern suggests that external conditions may be associated with marketing outcomes through activities extending beyond e-marketing adoption. Competition, changes in consumer preferences, institutional support, and access to business training may affect enterprise decisions concerning product presentation, customer service, pricing, and market positioning. Digital marketing represents one possible response to these conditions, but it does not encompass every aspect of how enterprises respond to their operating environment.

For food-processing MSMEs in Gianyar Regency, the distinction between direct and indirect environmental relationships is particularly relevant. Tourism-related demand may create opportunities for enterprises to reach new customers through digital channels, while competitive pressure may encourage more frequent online promotion. Government assistance and training may also provide resources that support digital adoption. However, the same external conditions may be associated with marketing performance through conventional business activities, including changes in product offerings or customer service arrangements. The positive direct and

indirect coefficients are consistent with this broader interpretation, although the present data do not establish the temporal sequence or causal mechanisms connecting these factors.

Comparison of the three indirect pathways reveals two related patterns. Technological context had the largest estimated indirect coefficient, while organizational context had the highest VAF percentage. These findings are not contradictory. The indirect coefficient describes the estimated magnitude of the pathway through e-marketing adoption, whereas VAF expresses that pathway as a proportion of the corresponding total effect. The organizational relationship therefore had a larger relative indirect component despite having a smaller absolute indirect coefficient than the technological relationship. The environmental context showed the lowest VAF, indicating that its estimated relationship with marketing performance was more strongly represented by the direct pathway. These differences provide a more detailed description of the structural relationships than a simple classification of all three dimensions as positive predictors.

The mediation findings also have implications for interpreting the TOE framework in the context of small food enterprises. Technological, organizational, and environmental conditions are often examined as determinants of technology adoption. The present analysis considers these conditions alongside marketing performance and separates their direct relationships from those estimated through e-marketing adoption. This distinction is relevant because the adoption of digital marketing tools represents an observable business practice rather than a general indication of technological readiness. The results suggest that enterprises reporting more favorable TOE conditions also tended to report more extensive digital marketing activities and stronger marketing outcomes. The estimated indirect pathways indicate that adoption may form part of the relationship between enterprise conditions and performance, subject to confirmation of their statistical significance.

From a practical perspective, the findings suggest that digital marketing initiatives for food-processing MSMEs should address both the conditions supporting adoption and the ways in which digital tools are used. Technological assistance may be relevant where enterprises encounter difficulties accessing or operating digital platforms. Organizational measures may focus on assigning marketing responsibilities, developing employee capabilities, and maintaining consistent customer communication. External support may be directed toward practical training and access to information about digital services. These areas are consistent with the factors identified in the structural model, although the present study did not evaluate the effectiveness of particular training programs or business interventions.

The findings should also be interpreted in light of the study design. The survey covered 94 enterprises selected through purposive sampling, all of which had experience using digital platforms. Consequently, the analysis concerns differences in adoption and marketing performance within this group rather than the decision to adopt e-marketing among all food-processing MSMEs in Gianyar Regency. The cross-sectional data also limit conclusions about the direction of the observed relationships. Enterprises with stronger marketing performance may possess greater resources for digital adoption, and the possibility of reciprocal relationships cannot be excluded. Furthermore, although the estimated indirect effects were positive, the available results do not provide the bootstrap confidence intervals needed to confirm the mediation hypotheses statistically.

Overall, the structural results indicate that technological, organizational, and environmental conditions were positively associated with both e-marketing adoption and marketing performance. E-marketing adoption also showed a positive relationship with marketing performance and accounted for part of the estimated total relationships between the TOE dimensions and performance. The analysis therefore identifies digital marketing adoption as a potentially relevant intervening pathway while recognizing that the contextual conditions of enterprises remain associated with marketing outcomes beyond that pathway. For food-processing MSMEs operating in tourism-oriented markets, the findings provide an empirical basis for considering technological access, organizational readiness, and external market conditions together when examining the adoption and use of digital marketing practices.

4. Conclusion

This study examined the relationships between technological, organizational, and environmental conditions, e-marketing adoption, and marketing performance among MSMEs producing animal-based processed food products in Gianyar Regency, Bali, Indonesia. Drawing on the TOE framework, the analysis considered both the factors associated with digital marketing adoption and the direct and indirect relationships between these factors and marketing performance. The findings provide empirical evidence from a segment of the food-processing industry in which traditional products, tourism-related demand, and digital marketing practices form part of the local business environment.

E-marketing adoption was found to be moderate among the surveyed enterprises. Although the participating businesses had already established a presence on digital platforms, including social media and online marketplaces, the reported extent of adoption indicated differences in how extensively these tools were incorporated into marketing activities. The findings suggest that the availability of digital services does not necessarily correspond to their consistent use for product promotion, customer communication, and market development. For small food

enterprises, the distinction between having access to digital platforms and integrating them into routine business practices remains important when assessing technological adoption.

The structural analysis identified positive and statistically significant relationships between all three TOE dimensions and e-marketing adoption. Technological context had the largest standardized coefficient ($\beta = 0.432$), followed by organizational context ($\beta = 0.372$) and environmental context ($\beta = 0.333$). These findings indicate that the accessibility, perceived usefulness, compatibility, and cost efficiency of digital technologies were particularly relevant to adoption within the surveyed sample. Organizational conditions, including management support, employee capabilities, digital experience, and infrastructure readiness, were also associated with the extent of adoption. External conditions, such as competitive pressure, customer expectations, community support, government assistance, and access to training, represented an additional dimension of the adoption process. Taken together, the results indicate that digital marketing adoption among food-processing MSMEs is associated with a combination of technological characteristics, internal organizational capacity, and the surrounding business environment.

E-marketing adoption was also positively associated with marketing performance ($\beta = 0.383$, $p < 0.001$), showing the largest direct coefficient among the predictors included in the performance model. Enterprises reporting more extensive digital marketing adoption also tended to report stronger marketing outcomes, as reflected in the performance dimensions examined in the study, including sales growth, customer development, customer loyalty, market expansion, promotional effectiveness, and brand equity. The technological, organizational, and environmental contexts retained positive direct relationships with marketing performance, indicating that these conditions were associated with marketing outcomes beyond their relationships with e-marketing adoption. The estimated indirect effects through adoption were 0.166, 0.143, and 0.128 for technological, organizational, and environmental contexts, respectively, accounting for 28.96%–37.53% of their corresponding total effects. These estimates suggest that e-marketing adoption may represent an intervening pathway linking enterprise conditions with marketing performance. However, confirmation of statistically significant mediation requires the corresponding bootstrap confidence intervals or significance tests for the indirect effects.

The study contributes to the application of the TOE framework by examining technology adoption alongside marketing performance within a specific segment of the food-processing sector. Rather than treating technological, organizational, and environmental conditions solely as determinants of adoption, the analysis distinguishes their direct relationships with marketing performance from the estimated indirect relationships operating through e-marketing practices. This distinction provides a more detailed account of how enterprise conditions and technology use are associated with marketing outcomes. The findings also extend the empirical application of the TOE framework to animal-based food-processing MSMEs operating in a tourism-oriented local market, where digital platforms are used to present traditional products, communicate with customers, and maintain market visibility. The contribution lies in the examination of these relationships within a defined industrial and geographical setting rather than in the development of a new technology adoption theory.

From a practical perspective, the findings point to several areas that deserve attention in the digital development of small food enterprises. Technological support should address the accessibility and suitability of digital tools for routine business activities, particularly where owners and employees have limited technical experience. Organizational measures may focus on staff training, clearer responsibility for online marketing, and the coordination of digital promotion with customer service and product availability. External assistance from government agencies, business associations, universities, and platform providers may be directed toward practical guidance on online business listings, digital content preparation, customer communication, and the use of marketplace services. For enterprises operating in tourism-oriented markets, these activities may be particularly relevant to maintaining contact with customers who obtain product and business information through digital channels. These recommendations are derived from the observed relationships and should not be interpreted as interventions whose effectiveness was directly evaluated in the study.

Several limitations should be considered when interpreting the findings. First, the analysis was based on 94 enterprises selected through purposive sampling in Ubud and Sukawati Districts. The sample was restricted to businesses already using digital platforms, potentially introducing selection bias and excluding enterprises that had not adopted e-marketing. Consequently, the moderate level of adoption reported in this study describes variation among the surveyed users rather than the overall adoption rate of all food-processing MSMEs in Gianyar Regency. Second, the study was restricted to a single regency, and its findings may not be directly applicable to enterprises operating in regions with different tourism activities, market structures, technological infrastructure, or institutional support. Third, the cross-sectional research design limits conclusions about the direction of the observed relationships. Although the structural model identified statistically significant associations, it cannot establish whether greater adoption preceded stronger marketing performance or whether better-performing enterprises were more capable of adopting digital technologies. In addition, the available mediation results do not include the bootstrap confidence intervals required for a complete assessment of the indirect effects.

Future research may address these limitations by examining enterprises across a wider range of geographical areas and food-related industries, including businesses that have not yet adopted digital marketing. Probability-

based sampling, where a suitable sampling frame is available, would allow stronger population-level inference and reduce some of the selection concerns associated with purposive recruitment. Longitudinal research could examine how digital marketing practices develop over time and whether changes in technological readiness, organizational capacity, and external conditions precede changes in marketing performance. Further analysis of specific digital activities, such as online ordering, social media communication, and customer engagement, may also help distinguish the relationships associated with different forms of adoption. Such investigations would provide a clearer understanding of the circumstances under which digital marketing becomes an established component of small-enterprise operations.

Overall, the findings indicate that e-marketing adoption is associated with technological readiness, organizational capacity, and external market conditions, while its relationship with marketing performance suggests that the practical use of digital platforms is relevant to the commercial activities of small food enterprises. The results provide a basis for considering technology access, internal preparation, and external support together when developing digital adoption strategies for food-processing MSMEs in tourism-oriented markets.

Author Contributions

Conceptualization, D.A.A.S.B. and G.M.K.A.; methodology, D.A.A.S.B. and G.M.K.A.; software, D.A.A.S.B.; validation, D.A.A.S.B. and G.M.K.A.; formal analysis, D.A.A.S.B.; investigation, D.A.A.S.B.; resources, D.A.A.S.B. and G.M.K.A.; data curation, D.A.A.S.B.; writing—original draft preparation, D.A.A.S.B.; writing—review and editing, D.A.A.S.B. and G.M.K.A.; visualization, D.A.A.S.B.; supervision, G.M.K.A.; project administration, D.A.A.S.B. Both authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon reasonable request.

Conflict of Interest

The authors declare no conflicts of interest.

References

Andika, A., Jennifer, J., Huang, J. C., & Sebastian, J. C. (2021). Analysis of digital marketing adoption in Indonesian micro, small, and medium enterprises. *J. Manaj. Bisnis*, *18*(3), 308–328. https://doi.org/10.38043/jmb.v18i3.3173.

Aqham, A. A., Rajendra, L., Huda, H. I., Sudirman, B., & Muthohir, M. (2024). Pemanfatan digital marketing di UMKM Kampoeng Jadoel melalui media sosial. *J. Abdimas Indones.*, *2*(4), 57–63. https://doi.org/10.59841/jurai.v2i4.2084.

Ardaba Kory, G. A. P. & Sanica, I. G. (2022). Strategi pemasaran digital UMKM di Bali dalam meningkatkan penjualan di era new normal. *J. Ilm. Edunomika*, *6*(1), 268–276. https://doi.org/10.29040/jie.v6i1.4053.

Astuti, A., Mane, A. A., & Saleh, M. Y. (2023). Peran perkembangan teknologi digital terhadap strategi pemasaran dan distribusi UMKM Kota Makassar. *Indones. J. Bus. Manag.*, *6*(1), 175–180. https://doi.org/10.35965/jbm.v6i1.3788.

Chaffey, D. & Ellis-Chadwick, F. (2019). *Digital Marketing: Strategy, Implementation and Practice (7th ed.)*. Pearson.

Fifian, F. (2025). Penerapan digital marketing untuk meningkatkan strategi pemasaran usaha mikro, kecil, dan menengah. *J. Ekon. Manaj. Sist. Inf.*, *6*(5), 3642–3650. https://doi.org/10.38035/jemsi.v6i5.5346.

Hakim, M. R., Abdillah, K., Sitinjak, G. A., & Prasetyoningtyas, A. A. W. (2025). Strategi pemasaran digital UMKM kuliner di era media sosial. *J. Pengabdi. Masy. Pemberdaya. Inov. Perubahan*, *5*(5), 161–170.

Jadhav, G. G., Gaikwad, S. V., & Bapat, D. (2023). A systematic literature review: Digital marketing and its impact on SMEs. *J. Indian Bus. Res.*, *15*(1), 76–91. https://doi.org/10.1108/JIBR-05-2022-0129.

Kaban, R. F., Hania, I., Widjaja, H. S., & Lubis, Z. (2024). The effect of digital marketing on the performance and sustainability of food MSMEs in Jakarta during the COVID-19 pandemic. *Jemasi: J. Ekon. Manaj. dan Akunt.*, *20*(1), 17–27. https://doi.org/10.35449/jemasi.v20i1.748.

Kotler, P. & Keller, K. L. (2016). *Marketing Management (15th ed.)*. Pearson Education.

Meranti, I. D. I. (2020). Transformasi digital marketing bagi UMKM sektor kuliner sebagai upaya pemberdayaan ekonomi pasca pandemi di Kabupaten Banyuwangi. *Economic: J. Ekon. Huk. Islam*, *11*(2), 142–152. https://doi.org/10.59943/economic.v11i02.66.

Mulyanto, D. & Budi, A. P. (2025). Penerapan pemasaran digital dan kinerja UMKM: Dukungan faktor

lingkungan internal dan eksternal. *J. Bisnis dan Kewirausahaan*, *14*(2), 301–313. https://doi.org/10.37476/jbk.v14i2.5173.

Nofrisel, N., Setiawan, E. B., Arubusman, D. A., Octora, T. Y., & Kholdun, A. I. (2023). The role of digital marketing in boosting MSME marketing performance. *Int. J. Econ. Manag. Res.*, *2*(2), 21–35. https://doi.org/10.55606/ijemr.v2i2.92.

Nuraini, R. & Mardikaningsih, R. (2024). Pengaruh literasi keuangan dan lingkungan bisnis terhadap kinerja bisnis (Survei pada UMKM). *Ekon. Keuang. Investasi Syariah (EKUITAS)*, *6*(1), 24–29. https://doi.org/10.47065/ekuitas.v6i1.5217.

Priyanto, H., Najib, M., & Septiani, S. (2020). Faktor adopsi e-marketing dan pengaruhnya terhadap kinerja pemasaran UKM kuliner Kota Bogor. *J. Sist. Inf. Bisnis*, *10*(2), 235–244. https://doi.org/10.21456/vol10iss2pp235-244.

Putri, V. P. & Arifiani, R. S. (2022). Orientasi pasar dan budaya organisasi dalam mempengaruhi inovasi dan kinerja pemasaran. *Magisma: J. Ilm. Ekon. dan Bisnis*, *10*(1), 11–21.

Rahman, A. & Riofita, H. (2024). Pengaruh lingkungan pemasaran mikro dan makro terhadap kinerja bisnis. *J. Econ. Excell. Ibnu Sina*, *2*(4), 212–218. https://doi.org/10.59841/excellence.v2i4.2087.

Safitri, L. S., Rahayu, W. E., Purwasih, R., Sobari, E., & Destiana, I. D. (2022). Pengenalan digital marketing dan m-commerce pada pelaku UMKM keripik singkong di Kabupaten Subang. *J. Pengabdi. Pemberdaya. Masy.*, *6*(2), 343–347. https://doi.org/10.30595/jppm.v6i2.9114.

Supriyati, S., Bahri, R. S., Khoerunisa, A., Natansya, T. S. F., Nissa, R. S., & Fernaldy, K. (2024). Peningkatan kinerja penjualan olahan beku melalui pemasaran digital produk mendekati kedaluwarsa pada UKM Pasar Ciroyom. *J. Appl. Finance Account.*, *6*(2), 125–132.

Svoboda, O. (2024). *Click to Success: Digital Marketing Step-by-Step Guide for Beginners*. eWay-CRM. https://www.eway-crm.com/eWay-Book/eWay-Book%20-%20Online%20Marketing%20EN.pdf

Tuluki, A. N. P., Soegoto, A. S., & Soepeno, D. (2023). Pengaruh kinerja pemasaran dan kualitas produk terhadap perkembangan UMKM di Kelurahan Teling Atas pada masa pandemi Covid 19. *J. Ris. Ekon. Manaj. Bisnis Akunt.*, *11*(3), 869–879. https://doi.org/10.35794/emba.v11i3.49201.

Wahyudin, M., Sadhewa, O. F., Yuliando, H., Chen, C. C., & Tsai, K. M. (2023). Character and readiness to adopt digital marketing technology: A case study of small retailers in Yogyakarta Province, Indonesia. *Eng. Appl. Sci. Res.*, *50*(4), 343–349. https://ph01.tci-thaijo.org/index.php/easr/article/view/251137

Zainarti, Nurwani, Nurbaiti, Aisyah, S., & Agustina, D. (2024). Digital marketing dengan memanfaatkan social media dan e-commerce bagi komunitas UMKM wanita. *J. Inov. Has. Pengabdi. Masy.*, *7*(2), 343–362. https://doi.org/10.33474/jipemas.v7i2.20873.